

FAMAIL: Fairness-Aware Trajectory Editing as Data Augmentation for Imitation-Learned Mobility Models

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ABSTRACT

Demand models learned by imitation from urban mobility data inherit the demographic service inequity encoded in their demonstrations. We present FAMAIL, a fairness-oriented method for data augmentation that *edits*, rather than regenerates, a small slice of real taxi trajectories chosen by attribution, then upweights the edited demonstrations so the fairness survives training. The editor combines two bounded, gradient-guided mechanisms under one differentiable objective: *trim* relocates recorded pickups within over-served areas so their measured over-service falls, and *lift* reroutes drivers' final seeking minutes into under-served areas, with the effect on local supply included in what the objective optimizes. Every edit is confined to a two-cell radius, with a frozen ST-SiameseNet driver-identity discriminator as a realism guardrail. On two cities (Shenzhen; San Francisco), editing improves established fairness measures that the objective never optimizes — demographic parity, disparate impact, and the Theil index. Where demand-only editing improves such measures purely by reducing over-service, the supply channel opened by lift adds statistically significant distinct-taxi presence to under-served areas in both cities. Upweighted behavior cloning carries the fairness-metric gains into trained policies, and does so specifically through the edited demonstrations: neither a random-upweighting nor a most-fair selection control reproduces the transfer. On Shenzhen, a targeted demographic-oversampling baseline captures only part of the data-level gain despite fabricating 10.5% of the corpus, and its untargeted placebo degrades fairness outright.

KEYWORDS

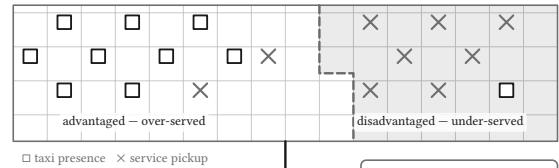
fairness, imitation learning, data augmentation, trajectory editing, urban mobility

1 INTRODUCTION

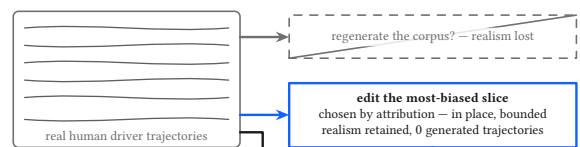
Demand models for urban mobility are increasingly learned by imitation: policies are trained to reproduce the trajectories a real fleet actually drove [33]. But trajectories carry more than behavior. They record the fleet's service allocation — which neighborhoods drivers sought passengers in, and which they passed over — and in real cities that allocation tracks who lives where: housing prices, incomes, migrant population share. A model that imitates such data faithfully learns the inequity as competence, and a model deployed to guide dispatch re-enacts it, sending supply where supply already went — the feedback loop documented for other data-driven allocation systems [6, 21] (Figure 1).

Interventions typically target one end of the pipeline or the other. In-processing methods regularize the model while leaving the demonstrations biased [34], so the objective and the training

(a) A real taxi fleet serves a divided city — with a 3.0× service gap



(b) The data encode valuable human expertise, which must be preserved



(c) Two corpora, two futures

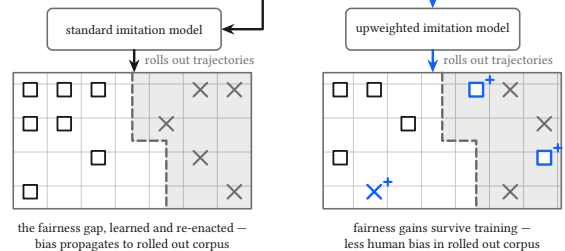


Figure 1: Why FAMAIL edits trajectory data. (a) A real fleet's service demonstrates bias: on Shenzhen the advantaged district receives 3.0× the taxi service per unit demand. (b) The recorded trajectories are human-generated and therefore valuable — filtering them would discard useful expertise, so FAMAIL performs bounded edits on an attributed slice of the corpus containing the most-biased demonstrations. (c) The stakes: a policy imitating the raw corpus learns the service gap and sends supply to over-served areas; trained on the upweighted edited demonstrations, the fairness survives training and results in the model learning a fairer policy. The + marks flag what the edit changed. Figure 2 shows how the editor works.

signal pull against each other at every step. Generative repair replaces the data wholesale, but rewriting an entire distribution risks trading realism for the fairness score and makes any improvement hard to attribute to anything in particular. FAMAIL takes a third position. As *data augmentation*, it keeps the real corpus, edits the small slice of it where attribution says the unfairness lives, and generates nothing. At most k trajectories are edited ($k = 10,000$ on the primary city, about a tenth of the corpus), each perturbed within a two-cell radius under a frozen driver-identity discriminator, so realism is inherited from the data rather than re-learned. The edited

demonstrations are then *upweighted* during imitation — instance reweighing [17] transplanted to demonstrations — so the fairness of a small slice survives training instead of being averaged away.

The subtlety is not measuring unfairness but repairing it honestly. FAMAIL's first editing mode, *trim*, moves only demand: it relocates recorded pickups within over-served areas, and on the primary city it improves measured fairness without helping the under-served group at all: every one of the 2,455 pickups it edits originates in an advantaged cell and not one lands in a disadvantaged cell, so the gap closes only from the top — analogous to the classic leveling-down objection to equalization [24, 26], though here service is relocated rather than removed. We show this behavior is structural — the constrained optimum of any demand-only editor on this objective (§3.4) — and that diagnosis dictated the second mode. *Lift* reroutes drivers' final seeking minutes into under-served areas and makes the supply consequences of the reroute differentiable and endogenous to the objective: the editor is rewarded for providing service where it improves fairness, not merely for redistributing demand. Together, trim and lift raise demand-adjusted demographic fairness on Shenzhen by +0.0226 and — for the first time in our experiments — add statistically robust taxi presence to the under-served group, raising its mean service ratio by +0.0176 under the optimizer's fractional-presence accounting (tier-1) and by +0.0411 when supply is recounted as distinct taxis from raw GPS (tier-2). On San Francisco the protocol reruns with no algorithm, metric, or fidelity change, and the data-level, external-metric, and downstream conclusions reproduce (+0.0316; §4.7).

Neither edit survives training on its own. The edited slice is a tenth of the corpus, and a vanilla behavior-cloning objective averages it away — a genuine null we verify rather than assume. Upweighting the edited demonstrations recovers the fairness with a monotone dose-response, and two controls pin the recovery to the edit itself: neither upweighting an equal-sized random subset nor upweighting the corpus's already-fairest trajectories reproduces it. Because a fairness method scored on its own objective proves little, evaluation rides on established measures the objective never optimizes — demographic parity, disparate impact (+0.0162 on the primary city's migrant axis), and the Theil index — computed before and after editing. On the primary city, a targeted demographic-oversampling baseline captures only part of the data-level gain despite fabricating 10.5% of the corpus, and its untargeted placebo degrades fairness outright.

Contributions.

- **A trim+lift trajectory editor.** Two bounded editing modes, gradient-guided under one differentiable objective, with the supply consequences of edits made endogenous to it, and driven by two exact attribution mechanisms: demand deficit attribution localizes existing unfairness, and supply-gradient attribution localizes the value of added taxi presence.
- **A structural diagnosis of demand-only editing.** Demand-only editing closes fairness gaps only from the top; we show why — selection, leverage, and frozen supply — and show that a differentiable supply channel is the non-perverse remedy.

- **A transfer recipe with edit-specificity controls.** Up-weighted imitation carries the data-level fairness into trained policies, and random-upweighting and most-fair selection controls isolate the edit as the cause.
- **Two-city validation on never-optimized measures.** The full protocol runs on Shenzhen and San Francisco and is scored on external fairness measures the objective never optimizes, against a demographic-oversampling baseline on the primary city.

2 RELATED WORK

Fairness interventions in machine learning. Algorithmic fairness interventions are conventionally grouped by where they act: pre-processing methods repair the training data, in-processing methods constrain the learner, and post-processing methods adjust its outputs [2]. FAMAIL belongs to the pre-processing family, whose canonical instruments — instance reweighing [17] and disparate-impact repair of feature distributions [7] — were built for labeled classification data. Among the many formal fairness definitions [30], the notion FAMAIL targets is conditional statistical parity [4]: disparities count only after permitted factors are adjusted for. Both instruments reappear here in unfamiliar roles — reweighing applied to demonstrations rather than labeled examples, and the parity condition transplanted from a classifier's predictions to a continuous spatial service ratio and made a differentiable editing objective.

Fairness in urban transportation. Transportation equity has its own measurement tradition, in which the Gini index over service distributions is the most widely used instrument [14, 18], alongside entropy- and welfare-based indices [1, 5, 28]. On the modeling side, the closest neighbor to this work regularizes a ride-hailing demand predictor in-processing, shrinking a racial prediction-error gap from 0.361 to 0.084 [34]. FAMAIL addresses the same demand-model bias one stage earlier — on the demonstrations the model trains on, rather than inside its loss — and is scored on equity measures its objective never optimizes.

Imitation learning for mobility, and mobility as identity. Imitation learning is an established route to urban mobility models: conditional GAIL variants recover taxi-driver strategies from trajectories [32, 33], explainable extensions expose the recovered policies [25], and neural simulators generate human mobility wholesale [8]. That literature optimizes realism; the demographic fairness of the imitated allocation is not among its objectives. A parallel line establishes that individual mobility is highly identifying: trajectory-user linking recovers who generated a trace [10, 23, 35], and Siamese verification decides whether two seeking traces share a driver [27]. FAMAIL contributes no new generator or imitation algorithm; it edits the demonstrations such models consume, and it repurposes the identity-verification line as a frozen realism guardrail — avoiding the instability of a live adversarial game [12].

Adversarial perturbation and recourse. The editing machinery descends from adversarial examples: bounded signed-gradient perturbations [11, 19], instantiated spatio-temporally for mobility signatures by ST-iFGSM [15], with discrete structure bridged by

continuous relaxation [16, 22] and straight-through gradient estimation [3]. Algorithmic recourse established that the same perturbation tooling built for counterfactuals can prescribe beneficial change rather than attack [29, 31] — for one individual’s feature vector, to flip one decision. FAMAIL extends that constructive reading from the individual to the population: the bounded perturbation edits real trajectories to reduce demographic service inequity, and the ε -bound is reinterpreted from an attack-stealth budget to a budget for preserving driver identity.

Leveling down and feedback loops. Two critiques frame what a fairness gain should mean. The leveling-down objection — equality achieved by making the better-off worse rather than the worse-off better — originates in the ethics of equality [26] and has been pressed concretely against machine-learning fairness constraints [24], with data augmentation observed to be the one intervention class that helps the disadvantaged group instead [36]. Feedback-loop analyses show that allocation systems trained on their own historical output entrench a self-reinforcing allocation bias [6, 21] — in our domain, the under-service of the neighborhoods that were under-served to begin with. Both critiques are load-bearing in this paper: we show our own demand-only editor closes gaps only from the top — an analogue of leveling down (§3.4) — build the supply channel expressly to lift up, and state demand endogeneity as the bound on what any demand-adjusted metric can see.

3 METHODOLOGY

3.1 Problem Formulation

We study taxi service allocation on a city discretized into 0.01° grid cells g and hourly time blocks t ; a pair $i = (g, t)$ with recorded activity is an *active unit*, and N denotes their count ($N \approx 34,500$ on the primary city). Each active unit carries a demand D_i — its recorded pickups — and a supply S_i — active-taxi presence, aggregated over the 5×5 cell neighborhood to match how a cruising taxi is visible to nearby demand. Service received relative to demand is the *service ratio* $Y_i = S_i / \max(D_i, d_0)$, where the demand floor $d_0 = 0.5$ prevents near-empty cells from producing unbounded ratios.

Each cell additionally carries demographic covariates x_i describing its neighborhood: housing price, per-capita compensation, and migrant population share. These covariates resolve at district granularity — about ten distinct profiles on the primary city — which bounds the resolution of any demographic analysis built on them (§3.2).

The data itself is a corpus $\mathcal{T}$ of real per-driver trajectories, each a sequence of passenger-seeking states ending in a pickup. The trajectories, not the aggregate grids, are the editable objects: D and S are deterministic aggregations of $\mathcal{T}$, so moving where a driver seeks and picks up moves demand and supply with it.

Task. Given a corpus $\mathcal{T}$ whose service allocation is demographically inequitable, produce an edited corpus $\mathcal{T}'$ that (i) is measurably fairer; (ii) remains realistic at the level of individual trajectories — edited trajectories must remain recognizable, in aggregate, to a frozen driver-identity discriminator as their original drivers; and (iii) transfers its fairness into a policy trained on it by imitation. FAMAIL is a *data-augmentation* method: it edits a small, attribution-targeted slice of the real trajectories — at most k of them ($k = 10,000$ on the primary city), each perturbed within an L_∞

ball of radius $\varepsilon = 2$ grid cells — and generates nothing. The edited slice is then upweighted during downstream training so that its fairness is not averaged away (§3.6).

3.2 The Fairness Objective

FAMAIL edits trajectories by gradient ascent on a single differentiable objective that combines demographic fairness, spatial equity, and realism.

Demand-adjusted demographic fairness (F_{causal} , primary).

Raw parity across neighborhoods is the wrong target: taxi service legitimately tracks demand, and a business district *should* see more pickups than a quiet residential block. We therefore measure fairness only in the component of service that demand does not explain — the logic of conditional statistical parity [4], which restricts disparities to those not explained by permitted factors. Stage one fits a flexible power basis $g_0(D)$ from per-unit demand to the service ratio and takes the residual $R_i = Y_i - g_0(D_i)$; stage two regresses that residual on the demographics and asks how much they explain. We evaluate stage two in closed form rather than through an explicitly fitted regression: with X the z-scored demographic matrix, $\tilde{X} = [1 \ X]$ its design matrix, $H = \tilde{X}(\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top$ the *hat matrix* of that regression — the projection onto the column space of $\tilde{X}$ [13] — and M the centering matrix,

$$F_{\text{causal}} = \frac{R^\top (I - H) R}{R^\top M R} = 1 - r_{\text{demo}}^2. \quad (1)$$

Despite its appearance, Eq. (1) is nothing but the stage-two regression computed in closed form: the numerator is its residual sum of squares, the denominator its total sum of squares. The projection form is what makes the measure usable as an *optimization objective* rather than a one-shot audit. Demographics are fixed attributes of the cells, so H is a constant of the optimization: as editing moves R , Eq. (1) implicitly re-fits the demographic regression exactly at every step, with no coefficients to refresh and nothing trained inside the editing loop. As a ratio of quadratic forms in R it is smooth, so exact gradients flow from the fairness score through Y to the coordinates being edited, and the idempotence of its projections underwrites the exact per-unit attribution of §3.3. Nor is the $N \times N$ projection ever materialized: by the normal equations, $R^\top (I - H) R = R^\top R - (\tilde{X}^\top R)^\top (\tilde{X}^\top \tilde{X})^{-1} (\tilde{X}^\top R)$, so each evaluation costs a few length- N inner products. By the Frisch-Waugh-Lovell theorem [9, 20], the residualize-then-project construction recovers exactly the demographic coefficients of the full multiple regression, so r_{demo}^2 is interpretable as the share of demand-adjusted service inequity attributable to demographics; measuring fairness through the statistical dependence between protected attributes and outcomes follows the predictability logic of [7], with the direction reversed: there, disparate impact is certified by predicting the protected attribute from the data; here, the demographics predict the adjusted service residual. Boundary cases orient the scale: $R \in \text{span}(X)$ gives $F_{\text{causal}} = 0$ (fully unfair), and $R \perp X$ gives 1 (fully fair).

Two caveats are load-bearing. First, F_{causal} is an *associational* quantity — the partial R^2 of a cross-sectional regression on observational district-level demographics, with no identification and no counterfactual; the “causal” in the name is historical. With about ten district profiles the association is ecological, and we make no

individual-level claims. Second, adjusting for demand treats recorded demand as exogenous; but demand recorded where service has historically been thin is itself suppressed by that under-service. We return to this in §3.4, where the same assumption turns out to bound what a demand-only editor can achieve.

Spatial equity (F_{spatial} , **secondary**). Independent of demographics, service should also be spread evenly relative to the taxi supply actually present. We track two supply-normalized service rates per active unit — the *demand-service ratio* $\text{DSR}_i = D_i/S_i$ (pickups per unit of taxi presence) and the *arrival-service ratio* $\text{ASR}_i = A_i/S_i$ (dropoffs per unit of taxi presence), with A_i the unit's recorded dropoffs and D_i , S_i as in §3.1 — and score the inequality of each with the Gini index, an inequality measure widely used in transportation equity research [14, 18]. The Gini enters as a pairwise mean absolute difference, differentiable almost everywhere and hence usable inside the gradient objective:

$$\begin{aligned} \text{Gini}(x) &= \frac{\sum_i \sum_j |x_i - x_j|}{2N^2 \bar{x}}, \\ F_{\text{spatial}} &= 1 - \frac{1}{2}(\text{Gini}(\text{DSR}) + \text{Gini}(\text{ASR})). \end{aligned} \quad (2)$$

As with F_{causal} , the quadratic pairwise sum is never materialized: a sorted prefix-sum identity evaluates it in $O(N \log N)$ with identical gradients. Being demographic-independent, the term acts as a spatial smoothness regularizer complementing F_{causal} .

Realism guardrail (F_{fidelity}). An edit that improves fairness by producing trajectories no real driver would generate is useless for training a realistic demand model. Because individual mobility signatures are highly identifying — the premise of the trajectory-user-linking literature [10, 23, 35] — we score realism with a frozen driver-identity discriminator of the ST-SiameseNet family [27]: an edited trajectory that still reads as its driver has preserved human fidelity. The discriminator is trained once and never updated during editing, avoiding the instability of a live adversarial game [12]. We state its role honestly: under the small, bounded edits FAMAIL makes, its gradient with respect to the edit is near zero, so it serves as a realism *guardrail* rather than a driver of edits — verified with respect to the edited pickup cell, with identity-axis stability under both edit modes confirmed empirically (§4); distributional collapse is monitored separately at evaluation time with a Jensen–Shannon check over aggregate mobility statistics [8].

Scalarization. The three terms combine linearly,

$$\mathcal{L} = \alpha_{\text{sp}} F_{\text{spatial}} + \alpha_{\text{ca}} F_{\text{causal}} + \alpha_{\text{fi}} F_{\text{fidelity}}, \quad \alpha = (0.1, 0.8, 0.1). \quad (3)$$

The weights are selected *empirically*: we sweep the weight simplex (fidelity fixed at 0.1) with full editing runs, and score every sweep point not only on the optimized metrics but on the design-targeted and external metric classes as well — classes defined in §4.1. The sweep exposes an asymmetry the optimized metrics alone would hide: ΔF_{causal} is flat (within 0.001) across $\alpha_{\text{sp}} \in [0, 0.55]$, but the supply-channel lift-up of the under-served group declines monotonically as α_{sp} grows, and beyond $\alpha_{\text{sp}} = 0.2$ it is no longer significant under tier-1 accounting (the tier-2 recount stays positive and significant at every sweep point). We therefore select by a three-class criterion — maximize ΔF_{causal} subject to $\Delta F_{\text{spatial}} \geq 0$ and the

supply-channel lift-up remaining significant under both accounting tiers — which the adopted configuration satisfies as the frontier's best point. Two properties make the choice robust rather than delicate: the primary gain is insensitive to the weights over a wide range, and the fidelity gradient is dormant under bounded edits — the discriminator constrains, but does not steer, the optimization.

3.3 Attribution: Localizing the Deficit and the Remedy

The objective is optimized one trajectory at a time, so FAMAIL must decide which trajectories are worth a slot in the edit budget. It uses two attribution mechanisms, one per editing mode (§3.5). *Demand deficit attribution* locates where existing unfairness concentrates, and drives the demand-side mode; *supply-gradient attribution* locates where added taxi presence would most improve fairness, and drives the supply-side mode. The first attributes the deficit; the second, the remedy.

Demand deficit attribution (drives trim). Because M and $I - H$ are idempotent, the explained share r_{demo}^2 of Eq. (1) admits an exact per-unit decomposition,

$$r_{\text{demo}}^2 = \sum_i \frac{(MR)_i^2 - ((I - H)R)_i^2}{R^T MR}, \quad (4)$$

a mathematically exact partition of the fairness deficit across active units; a signed variant separates over- from under-served units. The trajectories whose pickups land in the highest-deficit units are the ones trim edits.

Supply-gradient attribution (drives lift). The second mechanism asks, of every active unit: *how much fairer would the city's service be with marginally more taxi presence here, at this hour?* The quantity that answers it is the gradient of the objective with respect to supply, evaluated at the city's current state. To obtain it, we give the objective an auxiliary *supply-perturbation* input $\Delta S \in \mathbb{R}^N$ — one entry of hypothetical added taxi presence per active unit — and have it evaluate supply as $S + \Delta S$ in every term of the objective where supply appears: the service ratio Y inside F_{causal} and the rate denominators of F_{spatial} . At $\Delta S = 0$ the objective's value is untouched; the input exists so that automatic differentiation with respect to it reads out $\partial \mathcal{L} / \partial S_i$ at the city's current supply ($\Delta S = 0$) — the marginal fairness effect of added presence at each of the $N \approx 34,500$ active units, all obtained from a single backward pass. (The same ΔS pathway later carries the real supply consequences of lift edits, §3.5.) The result is a *value-of-presence* map over the city: large where added presence would most repair the fairness terms — units starved of supply relative to their demand and demographic profile — and small or negative where presence is already abundant. The map's F_{causal} component has the closed form

$$\frac{\partial F_{\text{causal}}}{\partial S_i} = \frac{2}{R^T MR} \frac{((I - H)R)_i - F_{\text{causal}} (MR)_i}{\max(D_i, d_0)}, \quad (5)$$

against which the automatic gradient is verified.

To turn the map into candidates, the screen asks which drivers already pass close enough to high-value territory that a bounded detour would take them into it. Each trajectory's tail — its last ℓ seeking states and pickup — is rigidly translated by every nonzero

integer offset $\delta \in [-\varepsilon, \varepsilon]^2$, and each translation is scored by its linearized gain: the value-of-presence collected at the tail's new positions minus that at its old ones, each state weighted by the presence mass it carries and the map summed over the same 5×5 window through which presence spreads (§3.1). A trajectory's score is its best offset, and every trajectory long enough to carry a movable tail ($\approx 80k$ of the corpus's $\approx 95k$) is ranked by score — the screen thus favors trajectories that already nearly serve the right areas, bending real behavior rather than inventing it. The screen only *nominates*: every selected trajectory is subsequently re-optimized individually under the full objective (§3.5), which derives the actual move. In assembling the edit budget, trim's selection takes precedence and lift fills the remaining slots with positive-score nominees.

3.4 Why Demand-Only Editing Cannot Help the Under-Served

Relocating pickups is a demand-side lever, and on its own it improves the fairness metrics in a specific, troubling way: it redistributes service among better-served areas while the under-served group gains nothing. On the primary city, every one of the 2,455 pickups edited by the demand-only editor originated in advantaged cells and not one landed in a disadvantaged cell; the under-served group's service was untouched, and the demographic gap closed only from the top. The pattern is analogous to the classic leveling-down objection to equalization [26] — measured equality improves while the worst-off gain nothing — recently formalized for machine-learning fairness together with a prescription to level *up* instead [24]; notably, data augmentation has been observed to be the one intervention that helps the disadvantaged group rather than pulling others down [36]. The analogy is inexact in one instructive way: trim *relocates* pickups under conservation rather than removing service, so nothing is destroyed — yet the under-served group is not helped at all, which is precisely the objection's force.

The behavior is the constrained optimum of the demand-only problem, not an optimizer artifact, for three compounding structural reasons. First, selection: demand deficit attribution (Eq. (4)) concentrates in over-served, high-residual cells, so under-served cells are never edit candidates. Second, leverage: with supply frozen, $\partial Y_i / \partial D_i = -S_i / D_i^2$ makes padding demand into high-supply advantaged cells roughly $32\times$ more consequential for Y than removing it from starved ones — and 93% of disadvantaged units sit at the demand floor d_0 , where removing demand changes nothing. Third, the inequity itself is on the supply side — median taxi presence is 1.8 in disadvantaged cells against 17.6 in advantaged ones — and a demand-only editor cannot touch supply. A greedy oracle (an unconstrained search that upper-bounds what any editor could achieve) closes the remaining escape route: the only demand-side way to raise the under-served group's service ratio is to delete its recorded pickups, which is perverse.

The deeper cause unifies the editor's limitation with the metric's. Recorded demand is not exogenous: where service has historically been thin, demand is suppressed and latent — the feedback-loop pathology documented for other public-allocation systems [6,

21]. The same censored demand that hides inequity from a demand-adjusted metric leaves a demand-only editor nothing to move in under-served areas.

This sets up the editor's second mode. The one non-perverse lever is the numerator: $\partial Y_i / \partial S_i = 1 / \max(D_i, d_0) > 0$ everywhere — at the demand floor, $\Delta Y = 2 \Delta S$ — so raising under-served service requires adding taxi *presence*, that is, rerouting real seeking behavior into under-served cells.

3.5 The Trim+Lift Editor

Figure 2 illustrates the two editing modes on a stylized city grid.

Shared machinery. Both editing modes repurpose bounded adversarial perturbation [11, 19], in its spatio-temporal instantiation [15], *constructively* — in the spirit of algorithmic recourse [29, 31], where the same counterfactual-perturbation tooling that finds a minimal input change to flip a decision instead prescribes a minimal, bounded, beneficial change. Per iteration, the edit offset takes a signed-gradient step under a cumulative L_∞ clip,

$$\delta \leftarrow \text{clip}(\eta \cdot \text{sign}(\nabla_\delta \mathcal{L}), -\varepsilon, \varepsilon), \quad \delta_{\text{total}} \leftarrow \text{clip}(\delta_{\text{total}} + \delta, -\varepsilon, \varepsilon), \quad (6)$$

so no pickup ever moves more than $\varepsilon = 2$ cells from its recorded location, regardless of iteration count — the ε -bound reinterpreted as an identity-preservation budget rather than an attack-stealth budget. The discrete grid is bridged by temperature-annealed soft cell assignment [3, 16, 22] over the same 5×5 window used to aggregate supply, moving from broad early exploration to precise late assignment; the best iterate is kept.

Trim. The demand-side mode: pickups selected by demand deficit attribution relocate within the ε -ball, padding recorded demand into over-served, supply-rich cells so their measured over-service falls (§3.4). Its optimization path is exactly the demand-only editor's.

Lift. Trajectories nominated by supply-gradient attribution have their whole seeking tail translated: the pickup moves by the full offset δ , earlier tail states move by linearly tapered fractions $w_j = j/\ell$ ($0.25/0.5/0.75/1.0$ at $\ell = 4$), and the anchor state before the tail never moves — so the trajectory physically cruises through the under-served area before picking up rather than teleporting into it. The moved states carry their supply with them, differentially: each seeking state contributes $1/12$ of an hourly presence unit to each cell of its 5×5 neighborhood in the supply perturbation ΔS , and the objective evaluates supply as $\text{clamp}(S + \Delta S, s_0)$ with supply floor $s_0 = 0.1$. Supply is thus *endogenous*: the optimizer sees the supply consequences of its own moves and is rewarded for providing service where it improves fairness, not merely for redistributing demand. The fidelity term scores the actual rerouted tail at every iteration, and each completed edit updates a shared running state, so later edits optimize against the city as earlier edits left it.

Physical validity. The source data is preprocessed — following the pipeline of [33] — to satisfy a king-move rule: consecutive states move at most one cell per axis, $\max(|dx|, |dy|) \leq 1$. Edits should not be held to a lower standard than the data they edit; a bounded two-cell pickup move would silently violate the very rule the source trajectories are filtered on. Discretized edits therefore pass through an exact backward-reachability repair that returns

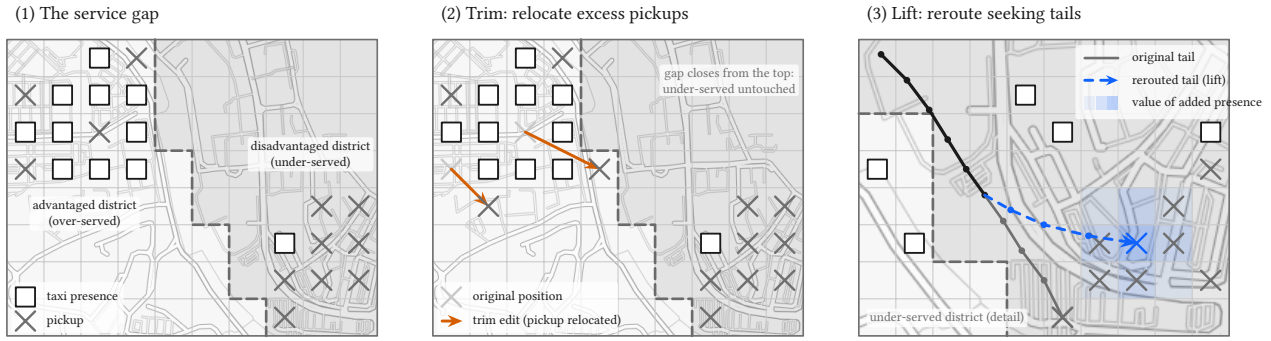


Figure 2: FAMAIL’s two editing modes on a stylized city: a street map discretized into grid cells, split by a district boundary into an advantaged and a disadvantaged district; pickups and taxi presence are quantized to cells. (1) The service gap: taxi presence concentrates in the advantaged district while the disadvantaged district carries unmet pickups. (2) *Trim* relocates excess pickups within the advantaged district — improving fairness metrics while the disadvantaged district gains nothing. (3) *Lift*, shown in a detail view of the under-served district: against the original tail, the rerouted tail bends the trajectory’s final seeking states into the cells where the supply gradient says added presence most improves fairness, carrying its taxi presence with it.

the king-compliant assignment nearest the tapered targets, or reports that none exists. Infeasible edits are not applied: lift skips them in-loop, and the small fraction of trim edits with no compliant repair (about 5%) is reverted to the originals post hoc under the same rule — a rule adopted before its metric effects were computed (§4).

Budget and phase order. Trim retains its selection by demand deficit attribution; lift fills the remaining budget with positive-score nominees — on the primary city, $k = 10,000$ splits into 2,455 trim and 7,545 lift edits — and the supply gradient is computed on the post-trim state. The two-phase structure is a scientific control, not an algorithmic necessity. Freezing trim keeps its optimization identical to the demand-only editor, which buys two properties: demand-only results reproduce exactly inside the combined run, and the trim-only versus trim+lift comparison is a clean ablation in which every difference is attributable to the new mechanism. A unified single-pass editor — supply endogenous everywhere, the gradient choosing each edit’s character — is a natural extension we leave as future work.

3.6 Downstream Recipe: Upweighted Imitation

Editing alone is necessary but not sufficient. The edited slice is small by design — roughly a tenth of the corpus — and a vanilla behavior-cloning objective, weighting every demonstration equally, averages it away and relearns the original bias; we verify this null empirically (§4). FAMAIL therefore *upweights* the edited demonstrations in the imitation loss — instance reweighing for fairness [17], a member of the pre-processing family alongside disparate-impact repair [7], transplanted to imitation learning. Whether the recovery is specific to the *edit*, rather than an artifact of oversampling or of selecting fair data, is tested against two controls — upweighting a size-matched random subset, and upweighting the corpus’s fairest trajectories — with outcomes reported in §4. This intervention targets, on the training data, the same demand-model bias that in-processing regularization has addressed at the model

level [34]; FAMAIL moves the intervention to the demonstrations themselves.

4 EXPERIMENTS

4.1 Experimental Setup

Datasets. The primary city is Shenzhen: a 50-driver sample of real taxi GPS trajectories on a 48×90 grid of 0.01° cells, with district-level demographics. Raw data contained per-driver “stuck-GPS” pickup sinks (single plates emitting thousands of phantom pickups); a signature rule removed 106,677 phantom pickups before any experiment. The external-validity city is San Francisco (Cabspotting): a 12-driver, fleet-density-matched subsample (sf12, 32×30 grid) with the same demographic feature names filled from ACS tract values. Density matching is required — the full 536-taxi fleet saturates the supply measure and leaves no editable fairness gradient.

Editor configuration. All edited corpora use the trim+lift editor of §3.5 with $\alpha = (0.1, 0.8, 0.1)$ — the configuration the three-class criterion picks on the measured frontier (§4.6) — $\epsilon = 2$, and the skip-on-infeasible rule. Shenzhen: $k = 10,000$ (2,337 trim + 7,545 lift after filtering); San Francisco: $k = 2,000$ (1,330 trim + 629 lift).

Protocol and statistics. Every model-based comparison uses paired seeds (identical seeding before each arm), so per-seed differences cancel shared seed noise. With $n = 6$ paired seeds the two-sided Wilcoxon signed-rank p floors at 0.03125, attained exactly when all six seeds share a sign; we therefore read effects from the mean Δ , its t -interval, and dose-response monotonicity, and treat $p = 0.031$ only as a sign-unanimity certificate. The flagship w30 comparison is additionally run at $n = 12$ paired seeds in both cities (floor .00049), where an exact Wilcoxon p can survive multiple-comparison correction. Suites with $n = 5$ seeds cannot reach $p < 0.05$ (floor 0.0625) and are reported as effect-versus-noise. Data-level before/after deltas on grid metrics carry paired

Table 1: Shenzhen external fairness, before → after editing (migrant axis, district extremes; paired bootstrap, $B=1,000$).

metric	before	after	Δ [95% CI]
Disparate impact ($\uparrow$)	0.3325	0.3487	+0.0162 [+0.0136, +0.0189]
DP gap ($\downarrow$)	14.199	13.309	-0.890 [-0.992, -0.785]
Theil ($\downarrow$)	0.1550	0.1463	-0.0087 [-0.0097, -0.0076]
mean(Y disadv.) ($\uparrow$)	7.0734	7.1263	+0.0529 [+0.0086, +0.0989]
mean(Y adv.)	21.272	20.435	-0.837

unit-level bootstrap intervals ($B = 1,000$, percentile); units are spatially correlated and demographics are district-constant, so these intervals are first-order.

Metric classes and claim discipline. We separate three metric classes and label every claim accordingly: (i) *optimized* — F_{spatial} , F_{causal} , F_{fidelity} , the quantities the editor’s gradient sees; (ii) *design-targeted but not optimized* — the family of supply/demand ratios, mean(Y) by group, which the lift mechanism was aimed at, so improvement there is confirmatory; and (iii) *genuinely external* — demographic parity, disparate impact, the Theil index, per-group service levels, and the distinct-taxi supply recount, none of which appear in the objective. “Improves metrics it never optimizes” claims ride class (iii) only.

External fairness instruments. On the service ratio Y over active units, grouped per equity axis (housing price, compensation, migrant share; two grouping conventions — district extremes and median split): the two group *levels* mean(Y | disadvantaged) and mean(Y | advantaged); *demographic parity* (DP), the signed gap between them (on a continuous outcome DP is the supply/demand-ratio gap — one number, reported once); *disparate impact* (DI), their ratio (1 = parity); and the between-region *Theil* index (grouping-independent). The strict count of distinct external instruments is therefore {DI, DP/gap, Theil} plus the descriptive levels.

4.2 Data-Level Fairness (Shenzhen)

Optimized metrics. Editing raises F_{causal} from 0.7988 to 0.8214 (+0.0226) and F_{spatial} from 0.1034 to 0.1095 (+0.0061) — class (i) numbers, reported for completeness rather than as evidence.

External metrics (class (iii)). Every established instrument moves toward fairness with bootstrap intervals excluding zero (Table 1, migrant axis, district extremes; the remaining axis $\times$ grouping cells behave the same way). The gap closes from both ends: the advantaged group’s over-service falls (21.27 → 20.44) while the disadvantaged group’s service *rises* (7.073 → 7.126).

The lift-up claim rides the supply channel. The mean(Y | disadv.) rise is a *design-targeted* (class (ii)) quantity — the lift mechanism was aimed at it — so we decompose it rather than celebrate it. Evaluating Y at $(S, D) \rightarrow (S, D') \rightarrow (S', D')$ splits the delta exactly into a demand move and a supply move (Table 2). The demand channel (pickup relocation) carries large variance and is not significant; the supply channel (rerouted tails adding presence) is significant under *both* accounting conventions — tier-1 (the optimizer’s fractional-presence convention) and tier-2, which re-counts supply from raw GPS as distinct taxis per neighborhood-hour with the

Table 2: Channel decomposition of Δ mean(Y | disadv.), Shenzhen migrant axis ($B=2,000$).

channel	Δ	95% CI	sig.?
Supply (tier-1)	+0.0176	[+0.0128, +0.0225]	yes
Supply (tier-2, distinct-taxi)	+0.0411	[+0.0346, +0.0479]	yes
Demand (pickup relocation)	+0.0352	[-0.0095, +0.0814]	no
Total	+0.0529	[+0.0086, +0.0989]	yes

Table 3: Trim-only versus trim+lift at matched budgets. CI = paired-bootstrap 95% interval; n.s. = interval includes zero.

	trim-only	trim+lift
<i>Shenzhen (primary)</i>		
ΔF_{causal}	+0.0146	+0.0226
$\Delta F_{\text{spatial}}$	-0.0011	+0.0061
Δ mean(Y disadv.)	+0.000 (flat)	+0.053 (CI excl. 0)
<i>San Francisco (external validity)</i>		
ΔF_{causal}	+0.0144	+0.0316
Migrant DP gap	-0.052 (CI excl. 0)	-0.073 (CI excl. 0)

edited tails substituted. The recount machinery reproduces the production supply grid exactly (MAE 0.0) with 100% history matching, so tier-2 is the stricter semantics, not an approximation.

Realism. Fidelity-A stability and lift’s expected Fidelity-B distributional cost (whole-tail relocation shifts trajectory-shape statistics more than a pickup-only move) are re-measured on the adopted corpus: Fidelity-A 0.844 vs. raw 0.848; corpus-level Fidelity-B 0.187, concentrated in exactly the components a rerouted tail must move — terminal cell (0.635), net displacement (0.147), coverage (0.142) — while trajectory length, which neither edit mode alters, barely registers (0.060) — a by-design trade-off we disclose rather than net out.

Provenance disclosures. About 5% of trim edits (118/2,455) had no king-compliant repair and were reverted post hoc; the rule was adopted before its metric effects were computed, and it moved every metric favorably — the infeasible edits were actively counter-productive. The pre-build oracle gate bounded the supply-channel headroom at +0.786 on mean(Y | disadv.) — $2.6\times$ the pre-registered go/no-go threshold of +0.3 (full accounting +0.882); the realized +0.053 sits far below the unconstrained ceiling, as a realism-bounded editor should.

4.3 Ablation: Trim-Only versus Trim+Lift

Because trim’s optimization is frozen inside the combined editor (§3.5), running the editor with the lift phase disabled reproduces the demand-only editor exactly, and every difference in Table 3 is attributable to the lift mechanism. The ablation certifies the design claim of §3.4: a demand-only editor improves fairness metrics while leaving the under-served group’s absolute service untouched — the gap closes only from the top — and the supply channel is what changes that.

Three rows carry the qualitative story, and the structural expectation from §3.4 re-verifies exactly at the adopted weights: under

Table 4: Data-quality comparison, four sources (Shenzhen; means over five seeds; the GAN row's Fidelity-B is seed-bimodal — see text).

source	F_{causal}	Fid.-A ($\uparrow$)	Fid.-B ($\downarrow$)
raw	0.7988	0.848	0.000
edited (trim+lift)	0.8214	0.844	0.187
bc-generated	0.7980	0.848	0.011
gan-generated	0.8089	0.848	0.171

trim-only, every external instrument still improves significantly (DP -0.611 , DI $+0.0098$, Theil -0.0059 , all CIs excluding zero) — but the disadvantaged group's service level is unchanged to four decimals ($7.0734 \rightarrow 7.0734$) while the advantaged level falls; the improvement is pure top-down redistribution: the under-served level does not move. Under trim+lift the same level rises with a bootstrap interval excluding zero. The secondary spatial metric flips from a small cost to a gain. Per Table 3, the lift mechanism is not an incremental tweak to the demand-only editor but the difference between redistributing service and providing it.

4.4 Downstream Propagation (Shenzhen)

Data quality (four sources). Table 4 scores the raw, edited, BC-generated, and GAN-generated corpora on fairness and both fidelity axes, with generators trained from scratch for this comparison and Fidelity-A gated by the real-anchored identity check (§3.2). The claim under test: the edited corpus is the fairest source that remains faithful on both realism axes.

The claim holds: the edited corpus is the fairest source and faithful to driver identity (Fidelity-A $0.844 \approx$ raw 0.848 ; gate passed, matched 0.848 vs. mismatched 0.193), and its Fidelity-B is the divergence of the bounded edits, quantified in §4.2. One instability we surface rather than smooth: the Shenzhen GAN's Fidelity-B mean of 0.171 hides a bimodal split across seeds (± 0.129 ; three of five seeds at 0.197 – 0.295 , two at 0.03 – 0.04), and its worst seeds fail broadband — length-component divergence 0.247 where the edited corpus sits at 0.060 . The collapse mode Shenzhen generators have shown before returns in some seeds here, while the SF GAN stays uniformly healthy (§4.7).

Vanilla training averages the edit away. A driver-conditioned BC policy trained on the edited corpus at uniform weight is not fairer than one trained on raw: paired $\Delta F_{\text{causal}} = +0.0016$ ($n = 12$, 7/12 positive, $p = .11$, n.s.) — the null that motivates the upweighting recipe of §3.6.

Upweighting recovers it, edit-specifically. With the edited demonstrations upweighted, the paired gain is monotone in the weight and sign-unanimous: $+0.0217$ (w10) and $+0.0267$ (w20), each 6/6 at the $p = .031$ floor. At the adopted dose we extend the flagship comparison to $n = 12$ paired seeds: $+0.0297 \pm 0.0029$ (w30), 12/12 positive, exact Wilcoxon $p = .00049$, t -interval $[+0.0278, +0.0315]$ — while both controls stay null at the same $n = 12$ (random -0.0017 , most-fair $+0.0012$, mixed signs, n.s.). Extending the dose shows saturation, not unbounded growth: $+0.0323$ (w40) and $+0.0339$ (w50), each 6/6, with the per-step increment shrinking monotonically ($+0.0050$, $+0.0035$, $+0.0021$, $+0.0016$) — w30 sits at

the knee of the curve, not at a tuned endpoint. Neither control reproduces it at any dose: the random-subset placebo is null-to-negative (-0.0014 w10, -0.0023 w30, -0.0036 w50, never positive), and the most-fair selection control reaches $+0.0033$ at w10 — significant, but under a sixth of the edited arm's gain — then fades to n.s. at every higher dose ($+0.0017$ w20, $+0.0009$ w30, $+0.0009$ at both w40 and w50). F_{spatial} propagates on Shenzhen as well: present already at uniform weight ($+0.0008$, $p = .031$) and rising through $+0.0038$, $+0.0040$, and $+0.0052$ across w10/w20/w30 (each $p = .031$), while at w30 both controls significantly *degrade* it (-0.0012 random, -0.0027 most-fair). Fidelity-A is unchanged at every weight (all n.s., $|\Delta|$ at most $+0.0007$); Fidelity-B rises dose-monotonically ($0.020 \rightarrow 0.027$ absolute vs. raw 0.013) — the expected signature of deliberately shifting population statistics toward the edited slice, disclosed rather than netted out.

Model-level variance. A paired baseline-vs-FAMAIL comparison at uniform weight gives $\Delta F_{\text{causal}} = +0.0030 \pm 0.0022$ at $n = 10$ seeds (9/10 positive; Wilcoxon $p = .0039$, t -test $p = .0019$) — the effect size is unchanged from the original five-seed suite ($+0.0031$) and now clears conventional significance outright. The reading stands: the training procedure alone transmits a small but real fraction of the edit, an order of magnitude below the upweighted gains, so the weighting lever remains what carries the transfer.

The allocation boundary. Metric propagation is not the whole story. Rolling out the trained policies and measuring the share of pickups they allocate to disadvantaged areas, the prior trim+lift corpus showed the share still tilting *away* — attenuated roughly 40% versus the demand-only corpus, but not reversed — and policies did not learn to *cruise* disadvantaged areas (seeking-state shares n.s.). This suite re-run on the adopted corpus — both sides, so the comparison is like-for-like at the adopted weights — gives the same boundary: the demand-only corpus tilts by -0.0049 at w30 and trim+lift by -0.0033 (each 0/6 seeds, $p = .031$) — roughly 33% attenuation, not a reversal — with seeking-state shares n.s. at every weight in both suites. We claim the data-level lift-up and the metric propagation; we disclose the allocation boundary, which motivates training-side allocation constraints as future work rather than further data-side editing alone.

4.5 Data-Augmentation Baselines

None of the baselines below optimizes a fairness objective; they are *editing-quality* baselines that test whether FAMAIL's gains come from its objective or merely from bounded perturbation (or corpus resampling) per se. Fairness is accordingly *expected not to improve* under the perturbation arms.

Perturbation arms. Three arms attack the same trajectories the headline edit selected, at the same $\epsilon = 2$: iterative and single-step signed-gradient attacks on the frozen identity discriminator, and a seeded random-jitter placebo. A naming note: the gradient arms are *iFGSM/FGSM with random restart* (PGD-style random initialization within the ϵ ball). We had pre-registered $\delta = 0$ -initialized ablation arms expecting a no-op — a difference-mode Siamese head has zero subgradient at an identical pair — but the deployed discriminator compares embeddings by *concatenation*, which is not stationary there: measured, the $\delta = 0$ arms attack rather than stall (the iterative arm lands within 0.0003 of its random-start twin; the

single-step arm differs with initialization, as a one-step method must). The stationarity assumption held only for the head we did not deploy, and the random-start arms are the reported ones. Results: Table 5.

The arms behave as fidelity baselines. One result lands against the pre-registered expectation: random jitter *does* raise F_{causal} , by +0.0135 — indiscriminately diffusing the selected trajectories diffuses their over-service too — but it does so by breaking the trajectories: 98.8% of its perturbed trajectories violate king-move adjacency and its distributional divergence reaches 0.447, against the edited corpus's disclosed 0.187 (iFGSM: 54.4%, 0.143; FGSM: 91.4%, 0.339; FAMAIL edits are king-compliant by construction). Identity fidelity survives every arm (≈ 0.846 , gate passed) — at $\varepsilon = 2$ the perturbations are too small to break driver identity, which is exactly why identity fidelity alone cannot rank editing quality here and the adjacency and distributional axes carry the comparison. The three perturbation arms replicate on San Francisco: every arm lands well below the editor (+0.0122 iFGSM, +0.0176 FGSM, +0.0049 random, against the editor's +0.0316) while violating adjacency wholesale (72.0/87.0/97.5%) and paying 2–4 $\times$ the edited corpus's distributional divergence (0.198–0.418 against 0.098); the random-jitter fairness surprise does not recur there — random is the weakest mover.

Demographic oversampling. The naive lift-up baseline: duplicate real trajectories originating in demographically disadvantaged regions (all three equity axes, phantom driver IDs, rigid ± 1 -cell jitter), additively rebuilding both demand and supply at doses matched to FAMAIL's budget. At the matched duplication dose $d = 10,000$, targeted oversampling improves F_{causal} by +0.0153 (dose-monotone: +0.0059, +0.0097, +0.0153) — but it fabricates 10.5% of the corpus to get there, where FAMAIL redistributes observed behavior at zero inflation and reaches +0.0226. The untargeted placebo (same fabrication without targeting) *degrades* fairness, -0.0172 , so the gain is targeting-specific; and the placebo's DP gap explodes (+2.8) because fabricated supply lands predominantly in already-disadvantaged cells — a measured demonstration that ratio metrics can be gamed by fabrication, the demand-endogeneity concern of §3.4 made concrete. Two disclosures: the corpus cannot even supply the matched dose without re-duplication (only 8,241 distinct disadvantaged-origin trajectories exist, so 1,759 draws repeat earlier ones), and fidelity is not scored for this arm — duplicates of real trajectories pass any realism check by construction.

Fairness-method baselines (training-side). The remaining comparison is against fairness methods proper, applied where they usually live: at training time, on the raw corpus, scored by the full paired protocol. Instance reweighing in the Kamiran–Calders style [17] — inverse group-service weights favoring disadvantaged-origin demonstrations — moves demand-adjusted fairness the *wrong* way (-0.0227 , all six seeds negative, $p = .031$) while paying the same distributional cost as w30 upweighting (Fidelity-B +0.0144), and its rolled-out policies still tilt allocation away from the disadvantaged group (-0.0010 , 0/6, $p = .031$). An in-processing penalty in the spirit of [34] — the imitation loss plus λ times the demographic-parity gap in predicted next-cell mass, tested in both the signed formulation (advantaged minus disadvantaged, the direction the data over-serves) and the absolute-value formulation — is inert at every dose that trains stably (six-seed suites: $|\Delta F_{\text{causal}}| \leq 0.0009$ signed

Table 5: Cross-arm comparison at matched budgets (Shenzhen).

arm	ΔF_{causal}	$\Delta F_{\text{spatial}}$	inflation
raw (reference)	—	—	0%
FAMAIL trim+lift	+0.0226	+0.0061	0%
iFGSM (rand. restart)	−0.0057	+0.0126	0%
FGSM (rand. restart)	+0.0017	+0.0137	0%
random jitter	+0.0135	+0.0132	0%
demog. oversampling (targeted)	+0.0153	+0.0052	10.5%
placebo (untargeted)	−0.0172	+0.0138	10.5%

at $\lambda \in \{1, 3.16, 10\}$, +0.0008 absolute at $\lambda = 10$, all n.s.; single-seed probes at $\lambda = 100$ agree; the term contributes on the order of 10^{-5} of the loss) and destructive at the dose where it finally dominates: at $\lambda = 1000$ the signed form collapses F_{causal} by -0.2053 with Fidelity-B at 0.514, the absolute form by -0.1293 (6/6 seeds) with Fidelity-B at 0.363, against raw's ≈ 0.013 . The shallower absolute collapse is the gap pushing back once it crosses zero; neither formulation offers a constructive operating range. Neither training-side intervention reproduces any part of the recovery that editing the demonstrations provides.

4.6 Robustness and Sensitivity

The control rows are re-measured deliberately: the random placebo is null-to-negative under every feature set, and the most-fair selection control — null on the primary set — is significantly *positive* under both alternate sets (+0.0054 and +0.0072 at w30, a fifth to a quarter of the edited arm's gain). We report the strongest controls we found rather than average them away: even where selection captures a minority share, the edited arm's gain is at least 3 $\times$ larger at every dose, which is the edit-specificity claim under its hardest test. Vanilla transfer likewise stays an order of magnitude below the upweighted gain under every set, reaching sign-unanimity only under the housing–GDP–compensation set (+0.0022, a tenth of its w30 gain).

Weight sensitivity. Fig. 3 reports the weight sweep: six full editing runs across the simplex, each scored on all three metric classes. The optimized ΔF_{causal} is flat — within 0.001 — for $\alpha_{\text{sp}} \in [0, 0.55]$, while the class (ii) supply-channel lift-up declines monotonically with α_{sp} — significant on both tiers only for $\alpha_{\text{sp}} \leq 0.2$ — and class (iii) instruments improve significantly at every point. The adopted (0.1, 0.8, 0.1) is the three-class criterion's selection; the sweep shows the choice is robust on the primary axis and load-bearing only for the lift-up property.

Demographic-feature-set robustness. F_{causal} is feature-set-specific, so every layer of the argument is re-measured under two alternate demographic sets (Table 6). The claim is directional reproduction, not magnitude equality — absolute F_{causal} scales shift with the feature set by construction. The design-targeted lift-up also reproduces: the total Δ mean($Y \mid \text{disadv.}$) carries a bootstrap interval excluding zero on all three sets, and the supply channel is individually significant on all three — under *both* accounting conventions (distinct-taxi tier-2: +0.0411, +0.0211, +0.0771, every interval excluding zero). The channel mix is not constant — on the primary set the demand channel is n.s., while under both alternate sets it

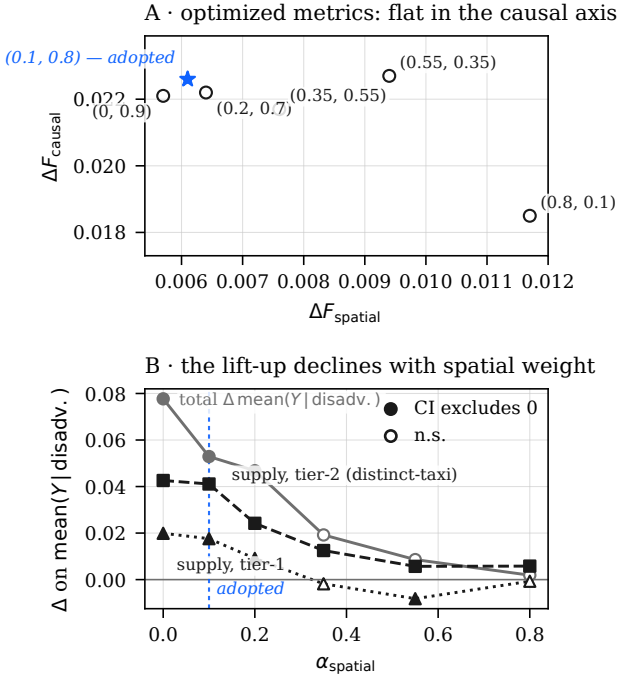


Figure 3: The weight sweep, scored beyond the optimized metrics. (A) On the optimized plane, ΔF_{causal} is flat across $\alpha_{\text{sp}} \in [0, 0.55]$. (B) The design-targeted lift-up declines monotonically with spatial weight and is significant on both supply tiers only for $\alpha_{\text{sp}} \leq 0.2$ (filled = 95% CI excludes 0). The adopted $(0.1, 0.8, 0.1)$ is the three-class criterion's selection.

Table 6: Feature-set robustness under trim+lift (Shenzhen; Δ rows are after – before edit).

	{hous,comp,migr}	{hous,gdp,comp}	+logpopdens
before-edit F_{causal}	0.7988	0.8069	0.7253
editor ΔF_{causal}	+0.0226	+0.0206	+0.0220
$\Delta \text{mean}(Y \text{disadv.})$	+0.0529	+0.0594	+0.1461
DI Δ (migrant/extremes)	+0.0162	+0.0147	+0.0191
DP gap Δ	-0.890	-0.787	-0.886
Theil Δ	-0.0087	-0.0080	-0.0085
L1: edited fairest faithful?	see Tab. 4	yes	yes
vanilla-BC transfer (w1)	see §4.4	+0.0022	+0.0011 (n.s.)
weighted-BC w30	see §4.4	+0.0248	+0.0256
random placebo (w30)	-0.0023 (n.s.)	-0.0019 (n.s.)	-0.0044 (n.s.)
most-fair select (w30)	+0.0009 (n.s.)	+0.0054	+0.0072
variance (b0 vs FAMAIL)	see §4.4	null	null

is significant and carries most of the larger totals – so the added-presence mechanism replicates while the demand-side share varies with the feature set, and we report the decomposition rather than the total alone.

Filtering is not a substitute. Removing the K least-fair trajectories instead of editing them does not reproduce the gain – it inverts it: F_{causal} falls monotonically with removal $(0.7988 \rightarrow 0.7935)$ at the maximum $K = 2,455$, while editing the same attribution-selected slice raises it to 0.8214. On the F_{causal} objective, neither

selecting (§4.4, most-fair selection control) nor removing data substitutes for editing it.

4.7 External Validity: San Francisco

The entire protocol reruns on San Francisco with no algorithm, metric, or fidelity change; because F_{causal} is city-specific and associational (ACS proxies, not hukou), San Francisco tests whether the conclusions reproduce, never whether magnitudes match.

Data level. F_{causal} rises $0.8752 \rightarrow 0.9067$ ($+0.0316$), F_{spatial} $+0.0139$, with Fidelity-A stability and lift's Fidelity-B cost re-measured on this corpus by the downstream suite (pending below). Every external instrument improves: Theil -0.0079 (CI $[-0.0094, -0.0065]$), and the migrant axis is significant under *both* grouping conventions on the DP gap (district extremes -0.0729 , CI $[-0.1132, -0.0259]$; median split -0.0370 , CI $[-0.0733, -0.0002]$), with DI significant under district extremes ($+0.0058$, CI $[+0.0004, +0.0103]$) and n.s. under median split – a grouping dependence we state rather than hide (the SF sample is 12 drivers).

The supply channel replicates; the ratio tension is a tier-1 accounting effect. The supply channel is again positive and significant under *both* conventions: $+0.0209$ tier-1 (CI $[+0.0122, +0.0300]$) and $+0.1027$ tier-2 (CI $[+0.0872, +0.1203]$) – the largest distinct-taxi supply effect of any corpus in this paper (Shenzhen's three feature sets: $+0.0411, +0.0211, +0.0771$; §4.6). But as on the prior corpus, lift also reroutes pickups into under-served cells, raising recorded demand there: the demand channel is significantly negative (-0.0533 , CI $[-0.0798, -0.0298]$), and under tier-1 accounting the total mean ($Y | \text{disadv.}$) moves net-negative (-0.0324 , CI excludes 0). We read the tier-1 net-negative as a demand-endogeneity finding, not a failed replication: recorded demand in under-served areas was suppressed, and serving it raises the ratio's denominator – the mechanism §3.4 predicts, observed in the wild. The tier-2 recount resolves the tension: counting supply as distinct taxis rather than fractional presence, the *total* turns significantly *positive* ($+0.0493$, CI $[+0.0185, +0.0790]$) – the real supply improvement outweighs the demand-side denominator increase. More rides served in under-served areas is a demand-side parity gain – DP, DI, and Theil all improve on San Francisco – while the supply channel independently replicates. The finer spatial partition plausibly amplifies the effect: San Francisco's demographic units are census tracts, far smaller than Shenzhen's districts, so the ratio is more sensitive to lift's rerouted pickups. A ratio-only reading confined to tier-1 accounting would withhold the SF lift-up claim; we report both accountings, and the stricter one supports it.

Downstream. The recovery reproduces on San Francisco. Vanilla BC is null ($+0.0017$, $n = 12$, 7/12 positive, n.s.); upweighting recovers the fairness with a monotone, sign-unanimous dose-response – $+0.0242$ (w10) and $+0.0309$ (w20), each 6/6 at the $p = .031$ floor – and at the adopted dose the flagship comparison reaches $n = 12$ here as well: $+0.0333 \pm 0.0050$ (w30), 12/12 positive, exact Wilcoxon $p = .00049$, as on Shenzhen (§4.4). The $n = 12$ controls sharpen the contrast rather than soften it: the random placebo turns significantly *negative* (-0.0077 , 2/12, $p = .0024$), and the most-fair selection control is significantly positive but dominated ($+0.0059$, 10/12, $p = .0020$, about a sixth of the edited arm's gain; at $n = 6$ per dose it was significantly negative at w10 and n.s. at w20).

The extended doses saturate here too (+0.0370 w40, +0.0371 w50, each 6/6 — flat past w40), while the random control stays null-to-negative and the most-fair selection control stays significantly positive (+0.0063/+0.0066) — the same dominated-control pattern as the alternate feature sets (§4.6), reported rather than averaged away. One city difference stated plainly: F_{spatial} does *not* propagate on SF (−0.0040 at w30, n.s.); the primary-city result at the adopted weights is reported in §4.4. The SF model-level variance suite stays null at $n = 10$ seeds ($\Delta F_{\text{causal}} = -0.0009 \pm 0.0035$, mixed signs, closer to zero than at $n = 5$) — unlike the primary city’s small significant positive effect (§4.4), a city difference we state plainly. The allocation boundary (§4.4) also replicates: rolled-out policies still tilt pickup share away from disadvantaged areas — −0.0432 at w30 on the demand-only corpus and −0.0335 with trim+lift (each 0/6 seeds, $p = .031$) — roughly 22% attenuation, not a reversal, at roughly ten times the primary city’s magnitudes. The four-source comparison also reproduces (generators retrained from scratch; identity gate passed, matched 0.958 vs. mismatched 0.034): the edited corpus is the fairest source (F_{causal} 0.9067 vs. GAN 0.8795, raw 0.8752, BC 0.8744) while identity-faithful (Fidelity-A 0.9581 $\approx$ raw 0.9578; Fidelity-B 0.098, the expected bounded-edit divergence).

San Francisco caveats. ACS foreign-born share proxies the migrant axis (not hukou); the 12-driver density-matched subsample is small; 14.9% of raw Cabspotting trajectories violate king-move adjacency at source (GPS gaps), so absolute compliance is unattainable — the edited corpus is *more* compliant than raw (87.6% vs. 85.2%), and the cross-city gate is edit-relative compliance (100%); and the GAN-generated source’s distributional collapse is a city-dependent property: the fresh SF run’s GAN stayed healthy, Fidelity-B 0.039, so any collapse-based cautionary note applies only to the city where it is measured.

5 CONCLUSION

FAMAIL makes fairness a property of the demonstrations. Rather than constraining a model or generating a synthetic corpus, it edits a small, attribution-targeted slice of real trajectories under one bounded, realism-guarded objective — trimming the measured overservice of advantaged areas, and lifting drivers’ final seeking minutes into under-served areas with the supply consequences of the reroute differentiable and endogenous — then upweights the edited demonstrations so the fairness survives imitation. On two cities the edits improve established fairness measures the objective never optimizes and, unlike the demand-only editor whose gains never touched the under-served group, the supply channel adds statistically significant taxi presence to that group; upweighted behavior cloning carries the fairness-metric gains into trained policies, and the control arms confirm the edit itself — not oversampling, not selection — is what carries them.

The claims come with stated bounds. The optimized fairness term is associational: a partial R^2 over roughly ten district-level demographic profiles, with no identification and no individual-level reading. Adjusting for recorded demand treats demand as exogenous, but demand is suppressed exactly where service has been historically thin, so a demand-adjusted metric under-detects the inequity it audits — the same censoring that bounds any demand-side editor. With five or six paired seeds, no single test survives

multiple-comparison correction; the one comparison extended to twelve paired seeds in each city — the flagship w30 recovery — does ($p = .00049$, 12/12 positive in both), and the remaining downstream evidence rests on effect direction and magnitude, confidence intervals, a monotone dose-response, and null controls rather than on any uncorrected p-value. with no algorithm change, but its baselines are city-specific and its magnitudes are not comparable. And the fidelity guardrail certifies that edited trajectories still read as their drivers — identity preservation, not fine-grained trajectory-shape realism.

Two directions follow naturally. The editor’s two-phase structure — trim first, then lift — is a scientific control, so the immediate extension is the editor §3.5 anticipates: a single pass in which supply is endogenous for every edit and the gradient decides each edit’s character. The second direction is transfer: whether other imitation objectives and adversarially trained generators inherit the edited corpus’s fairness the way upweighted behavior cloning does, whether allocation constraints on the training side can close the rollout gap that data editing alone only attenuates, and whether the recipe holds on further cities.

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